



RESTORATIONESSENTIALS

Factory-Correct Restoration Reference

LS/LT Conversion Guide — GM A-Body 1964–1972

Model Year 2026

Recipient cars	Chevelle/El Camino, GTO, Cutlass, Skylark/GS (1964–72 A-Body)
Donor engines	Gen III/IV LS (donor-first) · crate LS3 · Gen V LT
Highlights	13-chapter plan; gotchas named; worksheets included
Price	\$29.95 — PDF download at checkout

Every specification in this guide was checked against factory literature, marque references, period road tests, and documented survivor cars. Where sources disagree or the record is silent, this guide says so rather than guessing.

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Inside This Guide

The chapter order follows the RestorationEssentials house standard: history first, then year-specific identity, then the technical chapters you will use at the car.

- **Chapter 1: The Decision: Donor, Crate, or Gen V LT (Read This First)**



- **Chapter 2: What You Start With: The A-Body Baseline**
- **Chapter 3: Fitment: Putting an LS in a 1964–72 A-Body**
- **Chapter 4: The Gen V LT Chapter: Direct Injection Changes the Job**
- **Chapter 5: Wiring and Controls**
- **Chapter 6: The Fuel System: 58 Pounds of Non-Negotiable**
- **Chapter 7: Cooling and the Steam Lines**
- **Chapter 8: Exhaust: Headers, Manifolds, and the A/C Box Problem**
- **Chapter 9: ECU and Tuning Decisions**
- **Chapter 10: The Gotchas**
- **Chapter 11: Tips and Tricks**
- **Chapter 12: Special Tools**
- **Chapter 13: First Start and Shakedown**
- **Chapter 14: Appendix: Worksheets**
- **Chapter 15: Sources Consulted**
- **Chapter 16: Thank You**

Chapter 1 — The Decision: Donor, Crate, or Gen V LT (Read This First)

Every LS swap that goes wrong goes wrong here, before a wrench turns: the buyer prices an engine and discovers — in that order — the oil pan, the accessory drive, the fuel system, the wiring, the cooling, and the tune. This chapter puts the whole bill on the table first. Three roads lead into a 1964–72 A-Body (Chevelle/El Camino, GTO/Tempest/LeMans, Cutlass, Skylark/GS), and they are priced and risked differently.

- **Donor-first (Bill's approved path, and this guide's default):** a salvage truck 5.3 or 6.0 (LM7/LQ4) or an F-body/Corvette LS1/LS6 with its harness, PCM, and accessories. Cheapest entry; the unknowns are mileage, accessory completeness, and harness condition. The audit you do on the donor (compression, oil condition, accessory inventory) is Chapter 2 work.
- **Crate:** a new LS3 (430 hp from GM's published rating) or Connect-and-Cruise-type package with controller. Known quantity, real money, warranty. The swap hardware (pan, mounts, fuel, cooling) is the same bill either way — the crate buys certainty about the engine only.
- **Gen V LT:** the direct-injected LT1 6.2 (455 hp, GM's published rating) or truck L83/L86. Newer, more efficient, and a different job: high-pressure fuel hardware, VVT/AFM behavior, and Gen V accessory packaging. It has its own chapter (Chapter 4) because bolting it in like an LS is the expensive mistake.

The line items first-timers miss (planning ranges, not promises — prices move with salvage markets): the retrofit oil pan and pickup; the mount kit and transmission crossmember; the accessory-drive brackets if the donor drive doesn't fit; the EFI fuel system end to end (tank module or pump, lines, regulator, filters); radiator and fans sized for the new engine; the harness (reworked donor vs standalone) and the PCM edits (VATS delete at minimum); exhaust from the manifolds back; a driveshaft measurement and usually a new shaft; and the tune. If the total is beyond the budget, the honest alternative this guide endorses is the restoration guide for your car and a rebuilt original drivetrain — a finished car beats a stalled swap at every car show on earth.



Note. If you only have \$X: the money order that works is (1) the right oil pan and mounts, (2) the fuel system done once at full pressure spec, (3) cooling, (4) the PCM edits, and (5) cosmetics last. A swap that starts, cools, and charges on day one can be dressed for a decade. A pretty engine that overheats in traffic is yard art.

Chapter 2 — What You Start With: The A-Body Baseline

The recipient car gets inspected before the donor engine gets purchased. Baseline, with go/no-go criteria:

- **Frame:** the A-body's perimeter frame, checked at the steering-box mount, the front crossmember, and the rear coil-spring pockets. Cracks or section loss = repair first (the restoration guides cover frame work). The engine's torque didn't change the rules; it raised the stakes.
- **Steering:** identify the box and linkage (manual vs power, and which linkage generation). The 1964–67 cars place the engine more than an inch closer to the crossmember and steering linkage than 1968–72 cars — the documented reason early cars are the tight ones, and why your mock-up count (Chapter 11) is higher on a '65 than on a '70.
- **Firewall:** note whether the car has the factory A/C evaporator case. That plastic box on the passenger side of the firewall decides your header shortlist (Chapter 8) and your A/C future. Measure it; photograph it.
- **Electrical:** the car's 1960s wiring stays for lights and body; the engine gets its own harness and power feeds. Plan the battery and charging capacity for fans, pump, and ECM before the first start checklist (Chapter 13).
- **Ride height:** record it. Driveshaft length and pinion angle are measured at final ride height (worksheet A), and springs you plan to change change every number after them.

Chapter 3 — Fitment: Putting an LS in a 1964–72 A-Body

Mounts and position. Swap plate/mount systems place the LS from the stock small-block location to forward positions; the documented guidance is stock or slightly rearward for header and pan fit. On 1964–67 cars the tighter bay makes mount selection a clearance tool: adapter plates plus slightly taller mounts (the documented quarter-inch-taller solid or poly mount trick) buy pan-to-linkage room on early cars. Decide rubber, polyurethane, or solid by use — solids transmit vibration a street car doesn't need; the choice is recorded in the worksheet, not discovered at the first red light.

Oil pan — the A-body's first gotcha, handled here instead of Chapter 10 because it decides the mount conversation. Stock F-body and truck pans foul the A-body crossmember and steering linkage. The market's answer is the cast retrofit pan: Holley's 302-2 is listed for 1964–72 A-body applications, with a 3.62-inch maximum crank stroke (no LS7, no big strokers), a 5.7-quart sump, its matched pickup, an F-body windage tray (GM 12558253, with the tray's front quarter trimmed per the instructions) , and a specific dipstick/tube (Holley 302-15 or the LS3 pieces, GM 12669528/12625031). MEASURE ON YOUR CAR: pan-to-linkage clearance at full steering lock, with the engine at its final mount height — Holley's own documentation warns the engine position may need up/down or fore/aft adjustment depending on steering configuration.

Accessory drive. Three balancer offsets exist (Corvette closest to the block, F-body/GTO mid, truck tallest/furthest), and the donor's drive decides your bracket shopping: Holley's spacer kits (21-1 Corvette, 21-2 F-body/GTO, 21-3 truck/2010-up Camaro) exist precisely because one bracket set must serve all three offsets. The A-body frame rail and the truck drive's height are the classic conflict; the Corvette drive is the compact answer and the priced one.

Transmission tunnel and crossmember. The A-body tunnel swallows the period automatics and the 4L60E without drama; the 4L80E, 6L80, and the big-case manuals (T56/TR-6060/TKX conversions vary) get



MEASURED, not assumed — worksheet B. Aftermarket swap crossmembers (Hooker BlackHeart and kin) support the overdrives and set the driveline angle; the documented caution is body-bushing height: with half-height bushings, the crossmember's arches can need relief pockets in the floor and bellhousing clearance work. That is a permanent-modification decision; Chapter 10 grades it as one.

Chapter 4 — The Gen V LT Chapter: Direct Injection Changes the Job

The Gen V LT (LT1 6.2, 455 hp published; L83 5.3 and L86 6.2 in trucks; LT4 supercharged at 650 hp) bolts to the same mounts and brings four differences that change the plan:

- **Direct injection:** a cam-driven high-pressure pump feeds the rails; the in-tank/low-pressure side still has to be right, and DI lines and fittings are not hose-clamp territory. Fuel-system discipline from Chapter 6 applies double.
- **VVT and AFM:** variable valve timing and cylinder deactivation are built into the Gen V calibration. Keeping, locking, or deleting AFM is a tuning decision with hardware consequences (lifters), made in Chapter 9 — not discovered at the first oil change.
- **Accessory packaging:** Gen V accessory drives and the vacuum pump arrangement differ from LS; the swap-bracket market addresses LT separately. Do not buy LS brackets for an LT.
- **Electronics:** Gen V controllers (E92 family) and their security are a different programming conversation than the Gen III/IV PCMs. The standalone-harness decision (Chapter 5) is usually easier on an LT, and this chapter says so plainly.

Who should choose the LT: the buyer who wants the newest engine, accepts the electronics depth, and budgets the control package as part of the engine price. Who shouldn't: the first-timer on a tight budget — the Gen III truck donor path in Chapter 1 exists for him, and it finishes.

Chapter 5 — Wiring and Controls

The three harness strategies. (1) Reworked donor harness: cheapest, requires the discipline to thin it correctly and the PCM edits below; the donor's fuse/relay architecture comes along or gets rebuilt deliberately. (2) Standalone swap harness (PSI, Speartech, Holley and others): labeled, fused, and built for the swap; the budget line that buys weekends back. (3) Full engine-management systems (Holley Terminator-class) when the build outgrows the factory PCM's brief — cam, boost, or features the stock computer shouldn't be asked to fake.

The PCM edit list every factory-computer swap needs (documented across the swap literature): VATS/passkey security deleted — without it the PCM will not run the fuel pump, coils, or injectors, and the engine will crank forever in perfect health; rear oxygen sensors and the donor's emissions monitors addressed; tachometer output scaled to the classic gauge (the 4-cylinder/8-cylinder signal issue); vehicle-speed output scaled to the speedo solution you chose; electric-fan outputs enabled (truck PCMs did not drive electric fans from the factory); and the throttle decision — cable (early Gen III) or drive-by-wire with its matched pedal — programmed to match the hardware. Gauges: classic senders thread into LS ports with adapters; keep the mechanical temp sender honest and the oil-pressure sender matched to the gauge, not the donor.

Note. Safety: battery disconnected for every harness step; the ECM's grounds go to clean engine metal, not to a painted bracket. Ground-strap discipline is the modern ECU's favorite failure — a swap that 'has electrical gremlins' usually has three grounds sharing one corroded bolt.

Chapter 6 — The Fuel System: 58 Pounds of Non-Negotiable

The factory carburetor's mechanical pump and 3/8-inch line end here. LS EFI runs at a regulated 58 psi at the rail (published requirements cluster at 58–62 psi across MSD, FiTech, Edelbrock, and BluePrint documentation), fed by a pump flowing on the order of 240–340 liters/hour at pressure (BluePrint's published specification), through 30-micron-or-better filtration (100-micron pre-filter and a 10–30-micron post-filter around an inline pump, per the same documentation). The architectures:

- **EFI tank module:** a swap tank with an in-tank pump and regulator (the Tanks Inc.-style solution) — quiet, cooled by fuel, the clean-sheet answer and the one this guide recommends for a car being built once.
- **Inline pump with the stock tank:** works, with a sump/baffle plan for cornering starvation, the filter stages above, and a return line sized to the regulator's home (rail-mounted on early engines; filter/regulator combos on returnless-style builds).
- **Lines:** EFI-rated hose or hard line throughout, 37° AN flare fittings where the documentation calls for them, routed away from exhaust heat, with the tank vented deliberately — venting is a fuel-system item, not an afterthought (Chapter 10's fuel-smell gotcha lives here).

Note. Safety, plainly: 58 psi of gasoline finds every loose fitting at first prime. Prime the system with the coils unplugged, inspect every joint with a light and a dry hand, and keep a fire extinguisher — not a garden hose — within reach of the first start. Do not road-test a car with a weeping fuel fitting; there is no small gasoline leak.

Chapter 7 — Cooling and the Steam Lines

Sizing logic, not folklore: the radiator must shed the new engine's heat at idle in traffic with the A/C on if the car will have A/C — crossflow aluminum radiators and electric fans with a shroud are the market's answer, and the fan decision is electrical (Chapter 5's fan outputs and Chapter 2's charging capacity) as much as it is airflow. The swap-specific item is the steam-vent system: LS heads trap steam at the high corners, and the factory vent lines exist for that reason. Plumb the vents to the radiator or the pressurized side of the system per your radiator's provisions; an LS that 'runs hot only sometimes' is very often an unvented steam pocket telling you so. Diagnosis order for a hot swap: steam plumbing, fan trigger temperatures in the tune, shroud sealing, then parts — in that order, because the first three are free to check.

Chapter 8 — Exhaust: Headers, Manifolds, and the A/C Box Problem

The documented A-body trap pairs an engine-mount fact with a firewall fact: with swap mounts, stock low-mount A/C compressors are not installable — the compressor must upper-mount to the right cylinder head (Holley's 20-133 for the GM R4, 20-134 for Sanden SD508/SD7), and the Hooker/Holley documentation states the point in its own instructions. If the car will have A/C — aftermarket or factory-case — that decision precedes the engine going in, which is why the conversions family sequences its A/C guide where it does. On the exhaust itself: swap-specific headers exist for this chassis (Hooker BlackHeart mid-length and long-tube lines engineered with their mount system; Hedman mid- and full-length sets for adapter-plate installs, 1¾-inch primaries), and the same documentation records BlackHeart manifolds/headers as compatible with most stock A/C evaporator cases when used with the matching crossmember. Stock LS manifolds mostly do not fit the Chevelle frame — plan the header line item; where manifolds do clear, this guide says honestly what they cost in power: they lose some, clear everything, and never crack. Ground clearance with long-tubes on a lowered car is MEASURED ON YOUR CAR at final ride height.



Chapter 9 — ECU and Tuning Decisions

The decision tree, not brand loyalty:

- **Stock PCM, edited:** the Chapter 5 edit list, stock or near-stock engine — the cheapest correct answer, and the right one for a donor-fresh build.
- **Stock PCM, dyno tune:** required when the engine's airflow story changes — cam, heads, forced induction. A mail-order tune can delete security and set fans; it cannot know your cam. Anyone who tells you otherwise is selling, not tuning.
- **Standalone management:** when the build combines boost, transmission control, and features beyond the stock computer's brief, or on Gen V LT builds where the controller conversation is harder (Chapter 4).
- **Locked vs unlocked:** a swap PCM with security intact is a locked computer wearing a disguise. Verify the VATS status before first-start day (Chapter 13's checklist assumes it).

What a tune cannot fix: low fuel pressure (Chapter 6), steam-pocket cooling (Chapter 7), a slipping clutch, or a driveshaft angle (Appendix A). The tune is the last system finished, not the first excuse used.

Chapter 10 — The Gotchas

Each gotcha in the family's fixed format: *What happens / Why it happens / How to see it coming / The fix, in order of cost*. Categories the spec requires graded for this body are all here; where a category is clear on the A-body, it says so in one line.

- **Oil pan vs crossmember/steering.** *What:* the donor pan sits on the crossmember or kisses the center link at lock. *Why:* F-body/truck sumps were shaped for their own chassis. *See it:* mock-up with the linkage at full lock both ways (Chapter 3). *Fix, in cost order:* mount position adjustment; taller mounts on 1964–67 cars; the cast retrofit pan (302-2 class) with matched pickup and tray.
- **The low-mount A/C compressor.** *What:* the donor's low-mount compressor occupies the frame rail's space. *Why:* swap mounts put the engine where the compressor can't be. *See it:* Chapter 8 — decide A/C before the engine lands. *Fix:* upper-mount bracket (20-133/20-134 class); aftermarket A/C planned with the swap, not after.
- **The firewall evaporator case vs headers.** *What:* long-tubes meet the factory A/C case. *Why:* both want the passenger-side firewall. *See it:* Chapter 2's firewall measurement against the header maker's fitment notes. *Fix:* the documented-compatible manifold/header lines; case modification or deletion is permanent — decide knowing it.
- **Accessory drive vs frame rail (truck drive).** *What:* the truck accessory stack is tall and forward; the A-body rail and hood disagree. *Why:* balancer offsets differ by donor family. *See it:* Chapter 3's offset identification before purchase. *Fix:* bracket/spacer kits matched to offset; Corvette drive where clearance demands and budget allows.
- **Transmission tunnel vs the big overdrives.** *What:* 4L80E/6L80/big-case manuals can need floor relief, especially with half-height body bushings and arched swap crossmembers (documented). *See it:* worksheet B measurements. *Fix:* crossmember selection first, relief pockets last — permanent, so last.
- **Throttle-pedal geometry (drive-by-wire).** *What:* a DBW pedal mounted at the wrong angle gives a car that's lazy or touchy forever. *Why:* the PCM reads pedal angle, not effort. *See it:* mock the pedal position before final mount; verify full travel. *Fix:* reposition the pedal; no tune cures geometry.
- **Fuel-tank venting.** *What:* fuel smell, tank pressure, starvation. *Why:* the carbureted tank vented for 6 psi of drama, not 58. *See it:* Chapter 6's vent plan. *Fix:* vented cap/rollover vent routed correctly; charcoal canister where the build keeps one.



- **Donor accessory completeness.** *What:* the 'complete pullout' lacks the alternator, the pedal, the MAF, the PCM — each a separate purchase at retail. *Why:* 'complete' is a sales word. *See it:* Chapter 2's donor audit checklist, signed before money moves. *Fix:* buy the missing pieces as part of the negotiation, not after it.
- **Steam pockets.** Covered in Chapter 7 — graded here: applies to every LS/LT swap; vent the heads.
- **Driveshaft angle after ride-height change.** *What:* vibration that survives balancing. *Why:* lowering the car changed the working angles the shaft was built at. *See it:* worksheet A at final height. *Fix:* angle correction by shim/wedge per the drivetrain guides; a new shaft built to measured length — never to a forum's number.
- **Checked — clear on this body:** shock towers (the A-body has none to notch); header-to-steering-box clearance is a fitment-item per header line (Chapter 8), not a structural modification on the A-body.

Chapter 11 — Tips and Tricks

- **Plan on three engine installations:** mock-up, clearance-check, final. The swappers who do it once are the ones who planned to do it three times and got lucky twice.
- **Test-fit the accessory drive before final torque** on the mounts — belts and brackets are cheaper to rearrange on the stand.
- **Bag and tag everything** from the donor teardown; the harness connector you cut 'because it looked unused' is the one the PCM wants.
- **Chase threads before torque values** — 50-year-old frame threads lie to torque wrenches.
- **Photograph every routing decision;** heat and chafe are patient enemies, and the photo is the map at the first repair.
- **Ground straps:** engine-to-frame and engine-to-body, new, on clean metal. (Chapter 5's note is worth the repetition.)

Chapter 12 — Special Tools

Must-have: an engine hoist with a leveler (the leveler matters more than the hoist upgrade — the LS goes in at an angle, with the transmission's choice made), rated jack stands and a creeper you trust, a fuel-pressure gauge that reads 100 psi (Chapter 6's verification is not optional), a multimeter, and thread chasers for the frame's fastener sizes. **Rent or borrow:** the harmonic-balancer puller/installer, the transmission jack, the flare tool if hard lines are being bent. **Nice:** a lift — it converts the shakedown's under-car checks from an afternoon to an hour. No tool is listed here without the chapter that uses it.

Chapter 13 — First Start and Shakedown

- **Before cranking:** fuel system primed and leak-checked (Chapter 6's discipline); cooling filled and steam vents verified; oil primed per the engine's procedure; PCM security status confirmed (Chapter 9); battery charged; extinguisher in reach.
- **First start:** one person at the key, one at the engine bay with a light. Crank in short bursts; watch oil pressure like it's the only gauge (it is, today).
- **First heat cycle:** fan trigger verified against a real temperature reading; steam vents re-checked; every fastener you torqued gets its Chapter-13 re-check after cool-down where the service literature calls for it.
- **50 miles:** leaks, belts, harness chafe points, driveshaft vibration report (Appendix A if present). **500 miles:** re-torque list, fluid levels, alignment only after springs have settled — and the honest conversation about what the tune still owes you (Chapter 9).



Note. Safety, plainly: do not road-test the car until the brakes have been verified at walking pace and then at neighborhood pace — a swapped car has different weight and different pedal feel, and the first stop is a test, not a commute. Steering fasteners disturbed during the swap are single-use-critical: replace or re-torque per the factory literature, never 'snug by feel.'

Appendix: Worksheets (Fill These In On YOUR Car)

Worksheet A — Driveline. Ride height (four corners, as the car will be driven): _____. Transmission output angle: _____. Pinion angle: _____. Working angles computed: _____. Shaft length, center-to-center at ride height: _____. (The drivetrain family's shaft-and-geometry guide owns the targets; these blanks own your car.)

Worksheet B — Tunnel and clearance. Transmission chosen: _____. Bellhousing-to-floor clearance at closest point: _____. Crossmember: _____. Body bushings stock / half-height: _____. Relief pockets needed (Y/N, permanent): _____. Header-to-frame-rail clearance L/R: ____ / _____. Pan-to-center-link clearance at full lock L/R: ____ / _____.

Worksheet C — Parts-total ledger. Engine/donor: _____. Pan/pickup/mounts/crossmember: _____. Accessory drive: _____. Fuel system: _____. Cooling: _____. Harness/PCM/tune: _____. Exhaust: _____. Transmission and driveshaft: _____. Gauges and small parts: _____. Total, honest: _____. Compare it to the crate quote and the restoration-guide quote before Chapter 1's decision is final — that comparison is the decision.

Recommended for this restoration

Shop exact-fit supplies on Amazon

- LS swap A-body Chevelle motor mount kit
- Holley LS retrofit oil pan 302-2
- LS swap EFI fuel tank kit classic car
- LS standalone wiring harness

As an Amazon Associate, RestorationEssentials earns from qualifying purchases made through these links.

Sources Consulted

Every specification in this guide was checked against published references before printing. Where sources disagree, this guide says so in the text rather than picking a number silently. The references below are the backbone of the technical chapters:

- **Holley / Hooker published fitment documentation.** 302-series retrofit oil-pan specifications (stroke limit, capacity, windage-tray and dipstick requirements) and BlackHeart mount, bracket, and crossmember instructions including the low-mount compressor and body-bushing notes
- **GM LS-family production data.** engine RPO applications and factory ratings for Gen III, Gen IV, and Gen V engines as published in GM engine-family references
- **Marque swap references.** established LS-into-Chevelle swap literature (CarTech line, via chevydiy) for 1964–67 versus 1968–72 bay geometry and mount practice



- **EFI system manufacturers' installation literature.** fuel-pressure and filtration requirements as published by MSD, FiTech, Edelbrock, and BluePrint Engines
- **GM A-platform documentation.** 1964–72 A-body platform history, wheelbases, and division applications

A note on numbers. Factory literature, period road tests, and marque historians do not always agree — production totals, horsepower ratings, and paint codes all have documented conflicts. This guide uses the best-supported figure in each case and flags the conflict where one exists, so you can judge for yourself.

Emissions legality varies by jurisdiction: this guide explains what the factory-systems people in your state will ask; it cannot guarantee any conversion passes anywhere. Structural rust repair belongs to the restoration guides and to body shops; forced-induction beyond the factory-supercharged engines named here is outside this guide's scope.



Thank You

Thank you for choosing this RestorationEssentials guide. Whether this is your first LS/LT Conversion — GM A-Body 1964–1972 or your fiftieth, you picked one of the great ones, and the fact that you are doing your homework before you turn a wrench puts you ahead of most of the field.

These cars survived because people like you refused to let them disappear. Every correct fastener, every properly decoded tag, every hour spent getting the details right keeps the history honest for the next owner and the next generation of enthusiasts.

Modern engine, classic car — planned before the first wrench turned. Measure on your car, write it in the worksheets, and finish it: a finished swap beats a perfect plan every Saturday night.

If this guide helped your build, we would love to hear about it. Good luck with the restoration, and enjoy every mile.

— The RestorationEssentials Team